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Analyzing Crime Patterns and Forecasting Trends

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DS 4420: Machine Learning and Data Mining 2

Purpose

- Analyze crime data from Los Angeles (2020–2024) to uncover patterns and trends in criminal activity.
- To assist law enforcement in proactive resource allocation based on predicted crime hotspots.
- Explore how crime evolves over time and across different offense types.

Background & Data

- Crime in urban environments exhibits strong seasonal and temporal patterns, making it suitable for time series forecasting.
- Bayesian approaches provide a logical way to integrate previous information and measure uncertainty in crime predictions.

Feature	Description
Date/Time Occ	Date & time of incident
Area Name	Los Angeles Police Department Division
Crm Cd Desc	Crime type/description
Vict Sex/Descent	Victim demographics
Premis Desc	Location type (street, home...)
Weapon Desc	Weapon used
Status	Invest. Cont., Adult Arrest...

Method

Bayesian Modeling

- **Preprocessing:** sorting by time, adding new column for cleaner date, aggregating by date and area
- **Training data:** 2020-2023, testing data: 2024
- **Model:**

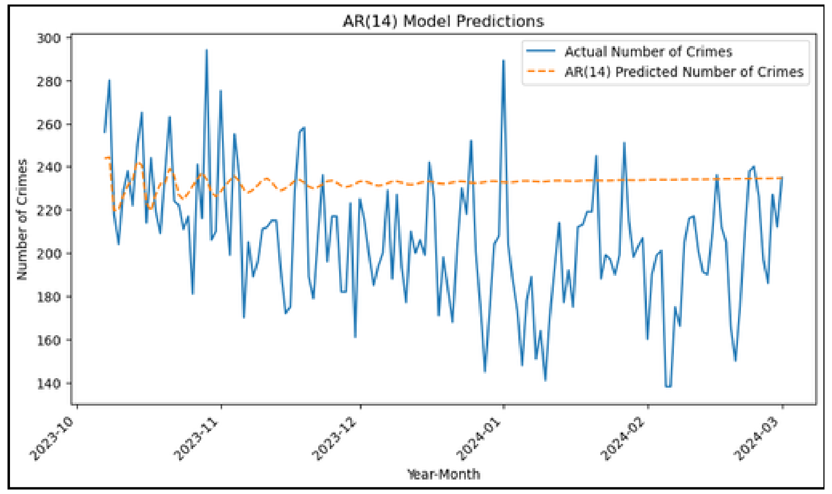
1. Predict the number of crimes for the whole Los Angeles using the Poisson distribution
2. Predict the number of crimes for each area using Poisson distribution

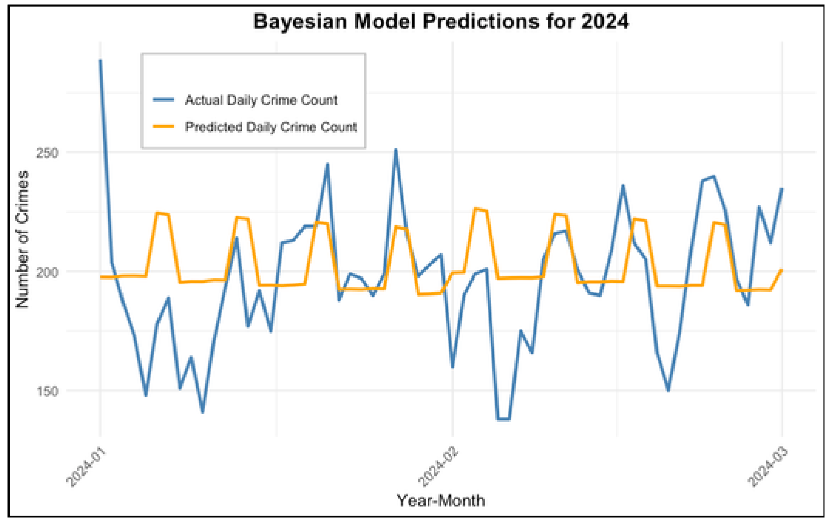
- **Evaluation:** Generated posterior predictive distributions for 2024 test data

Time-series Modeling

- **Dataset**: daily crime counts (Los Angeles, 2020–2024)
- **Preprocessing:** datetime conversion, sorting, aggregation
- **Analysis**: time series plot, ACF, PACF (p = 14)
- **Model**: AR(14) using lag features (1–14), least squares estimation
- **Prediction**: recursive forecasting using previous values
- **Evaluation**: 80/20 split and prediction comparison

Result





Time Series Model

- The AR(14) model captures the overall average trend in crime counts but produces overly smooth predictions.
- It fails to capture sharp fluctuations, indicating limited performance for highly variable crime data.

```
# Predict for specific area and time
# Example: Hollywood in February 2024
predictions_specific <- test %>%
  filter(AREA.NAME == "Hollywood", month(Date) == 2)
```

```
> mean(predictions_specific$Predicted)
[1] 12.3821
> mean(predictions_specific$Count)
[1] 12.03448
```

Bayesian Model

Predict the number of crimes for the whole of Los Angeles using the Poisson distribution

- The Bayesian model yielded a result quite similar to that of the Time Series model. The predicted values closely tracked the general trend of actual crime counts, maintaining estimates in the 190-225 range that reflected the underlying baseline activity.
- However, the model had difficulty capturing day-to-day fluctuations.

Predict the number of crimes for each area using the Poisson Distribution

- Using Hollywood in February 2024 as a test case, the model produced a posterior predictive mean of 12.38, compared to the observed mean of 12.03. This indicates strong predictive accuracy for that month.
- However, across other months, predicted means consistently converged toward the 11-12 range regardless of the true observed mean. This reflects the regularizing influence of the prior.

Conclusion

- Both the AR(14) and Bayesian models successfully predicted the general trend of daily crime counts in Los Angeles.
- However, both models struggled to reproduce daily or monthly fluctuations. This suggests that crime volatility is driven by factors not fully captured by time factors alone.

Future Work

- Explore multivariate time series to model crime type interactions across areas.
- Incorporate other socioeconomic factors into the model, including unemployment rates, poverty levels, housing instability, etc. to better understand the structural issues that drive crime.
- Adopt more complex model architectures to more accurately reproduce the fluctuations.

References

Crime Data from 2020 to 2024. City of Los Angeles (data.lacity.org). Published February 10, 2020; last modified March 4, 2026. Metadata updated March 8, 2026. U.S. Data.gov Catalog. https://catalog.data.gov/dataset/crime-data-from-2020-to-present data